Summary Report: Unsupervised Learning - Clustering Techniques

Project Overview

This project introduces the basic concepts of unsupervised learning using clustering algorithms. Three clustering techniques were implemented using the Iris dataset to understand how similar data points can be grouped without using labeled data.

Dataset Used

Dataset: Iris Dataset

Source: Scikit-learn Built-in Dataset

The dataset contains 150 flower samples with four numerical features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Algorithms Implemented

K-Means Clustering

K-Means groups data into a predefined number of clusters based on the distance between data points and cluster centroids.

Hierarchical Clustering

Hierarchical Clustering creates clusters by gradually merging similar data points into larger groups.

DBSCAN

DBSCAN groups data based on density and can identify noise or outlier points without specifying the number of clusters.

Conclusion

This project provided a basic understanding of clustering techniques in unsupervised learning. It demonstrated how K-Means, Hierarchical Clustering, and DBSCAN can be used to identify patterns and group similar data points using the Iris dataset.
